StockSense

Inventory, sales and smart reordering for small shops. Project overview, October 2026

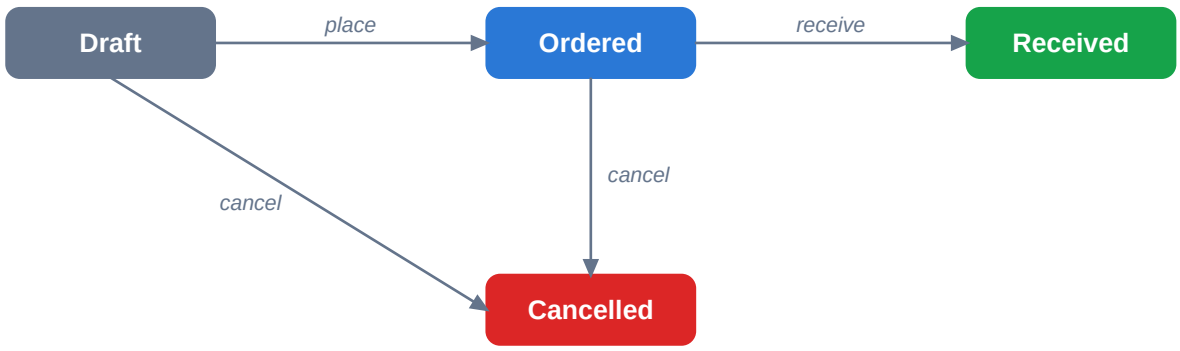
What it is

StockSense is a web application that helps a small shop keep the right amount of stock. It records products, sales and purchase orders, keeps a full stock ledger, and forecasts each product's demand from its own sales history to suggest **when** to reorder and **how much**. Every number can be traced back to the shop's data. No external AI service is used.

Modules

Module	What it does
Dashboard	Active products, low-stock items, sales today and in the last 30 days, open purchase orders, and reorder alerts.
Products, categories, suppliers	Catalog with SKU, cost and sale price, reorder level, and each supplier's lead time.
Sales	Records a sale and deducts stock. A sale that exceeds available stock is rejected, and nothing is saved.
Purchase orders	Status-based workflow from draft to received. Stock is added only when an order is received.
Demand forecasts	Daily demand forecast, stock-out date and suggested order quantity for every product.
Users, roles, permissions	Role-based access with fine-grained permissions, for example "receive purchase orders" or "view forecasts".

Purchase order workflow



- Only **draft** orders can be edited or deleted.
- **Place** needs at least one item. It records the order date and, if none was given, sets the expected date from the supplier's lead time.
- **Receive** locks the order and its products, adds each item to stock through the ledger, and records who received it. An order can't be received twice.
- A draft or a placed order can be **cancelled**. Any other move is rejected with a validation error.

Demand forecast and reorder suggestions

Each product gets its own self-trained model: **damped-trend exponential smoothing** with a learned **weekday pattern**, so busy days such as Fridays are forecast higher. The model learns from the last 90 days of sales.

Days when the product was out of stock are treated as unknown demand, not as zero demand.

- **Back-tested:** the model is trained without the last 14 days, asked to predict them, and compared with a 28-day moving average (WAPE), so its accuracy is measured, not assumed.
- **Reorder policy:** classic (s, S). When stock plus open orders falls to the reorder point (demand over the lead time plus safety stock), order up to the target (lead time plus a 7-day review period plus safety stock).
- **Safety stock:** covers demand on about 95% of order cycles.
- **Status:** *critical* (stock runs out before a new order could arrive), *reorder*, *ok* or *no data*.

Technology

Layer	Technology
Backend	Laravel 13 on PHP 8.4+, with a service layer per module and a versioned JSON API (/v1/admin)
Frontend	Inertia.js v3, React 19, TypeScript and Tailwind CSS v4, built with Vite 8
Database	PostgreSQL. Row locks keep sales and receipts safe under concurrent use.
Auth and access	Laravel Sanctum session auth, with roles and permissions from spatie/laravel-permission
Dev environment	Docker through Laravel Sail. Tested with PHPUnit and formatted with Laravel Pint.

Running it locally

Clone the repository, install dependencies, configure `.env` for PostgreSQL, then run `php artisan migrate --seed`. The seeder creates a Super Admin (`admin@stocksense.test` / `password`), sample products and suppliers, and 90 days of sales and purchase history, so the forecasts work immediately. See `README.md` for full steps.